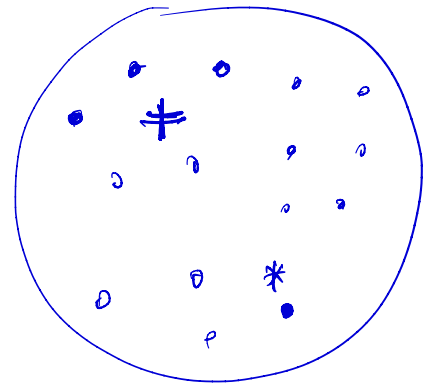


$$K \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} Ka_1 \\ Ka_2 \end{bmatrix}$$

Module 2



$$K \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} Ka_1 \\ 0 \end{bmatrix}$$

Chapter 3

Vector Spaces, Linear dependence and Bases

Vector Space:

Throughout this course, we study vectors in a Euclidean Space. That is vectors that can be represented as an array of numbers, like:

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

These vectors can be added together or scaled by multiplying by another scalar and we end up getting another vector. For example:

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

Or

$$2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

A **vector space** generalizes this notion of adding and scaling things that behave like Euclidean vectors.

A “**vector space**” consists of a set of vectors, as well as two operations: addition of vectors, and multiplication by scalar values. To be a valid vector space, the set of vectors must be closed under these two operations; that is, adding a multiple of any vector will result in a vector that is still in the space. The methods of vector addition and scalar multiplication must satisfy specific requirements/ axioms:

Let V be any non-empty set of objects. Let's define on V an operation, called *addition*, for any two elements belonging to V i.e. $\vec{x}, \vec{y} \in V$, and denote this operation by $\vec{x} + \vec{y}$. Let *scalar multiplication* be defined for a real number $a \in \mathbb{R}$ and any element $\vec{x} \in V$ and denote this operation by $a\vec{x}$. The set V together with operations of addition and scalar multiplication is called a vector space over $\mathbb{R}$ if the following hold for all $\vec{x}, \vec{y}, \vec{z} \in V$, and $a, b \in \mathbb{R}$:

- $\vec{x} + \vec{y} = \vec{y} + \vec{x}$
 - $(\vec{x} + \vec{y}) + \vec{z} = \vec{x} + (\vec{y} + \vec{z})$
 - There exists some vector $\vec{0} \in V$, such that $\vec{x} + \vec{0} = \vec{x}$
 - For each vector $\vec{x} \in V$, there exists a unique vector $-\vec{x} \in V$, such that $-\vec{x} + \vec{x} = \vec{0}$
- Handwritten examples illustrating the properties:
- $$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} -2a_1 \\ -2a_2 \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$
- $$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \end{bmatrix}$$
- $$-\begin{bmatrix} a_1 + (-2a_1) \\ a_2 + (-2a_2) \end{bmatrix} = -\begin{bmatrix} -a_1 \\ -a_2 \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

These are the main properties/axioms associated with the operation of addition.

The next four properties are associated with the operation of scalar multiplication and how it relates to vector addition.

- $a(\vec{x} + \vec{y}) = a\vec{x} + a\vec{y}$
 - $(a+b)\vec{x} = a\vec{x} + b\vec{x}$
 - $a(b\vec{x}) = (ab)\vec{x}$
 - $1\vec{x} = \vec{x}$.
- Handwritten notes: A blue bracket groups the first four properties. To the right, a blue vector $\begin{bmatrix} a \\ b \end{bmatrix}$ is multiplied by a scalar k to give $\begin{bmatrix} 0 \\ kb \end{bmatrix}$. Below the bracket, a red vector $\begin{bmatrix} a \\ b \end{bmatrix}$ is multiplied by a scalar k to give $\begin{bmatrix} ka \\ kb \end{bmatrix}$.

For example:

Let $\mathbb{R}^2 = \{(a_1, a_2) \mid a_1, a_2 \in \mathbb{R}\}$. If we define addition and scalar multiplication the natural way, that is, as we would on 1×2 matrices, then $\mathbb{R}^2$ is a **Vector Space over $\mathbb{R}$** .

Let's look at an example:

Let us see if a given set of vectors is a valid vector space or not:

Handwritten: $\begin{bmatrix} a \\ 0 \end{bmatrix} \longrightarrow \begin{bmatrix} a \\ 3a+1 \end{bmatrix}$

Let V be a set of vectors given by: $\{(a, b) \in \mathbb{R}^2 : b = 3a + 1\}$

We need to check if the set is closed under vector addition and scalar multiplication:

Let there be two vectors $\vec{x} \in V$ and $\vec{y} \in V$.

Let $\vec{x} = \begin{bmatrix} a_1 \\ 3a_1 + 1 \end{bmatrix}$ and $\vec{y} = \begin{bmatrix} a_2 \\ 3a_2 + 1 \end{bmatrix}$

Then, $\vec{x} + \vec{y} = \begin{bmatrix} a_1 + a_2 \\ 3(a_1 + a_2) + 2 \end{bmatrix} \notin V$

Handwritten calculation for scalar multiplication:

$$k \begin{bmatrix} a_1 \\ 3a_1 + 1 \end{bmatrix} = \begin{bmatrix} ka_1 \\ 3ka_1 + k \end{bmatrix} = \begin{bmatrix} A \\ 3A + K \end{bmatrix}$$

where $A = ka_1$ and $K = k$.

Hence, this is not a valid subspace!

Let's Practice

Is V a valid Vector Space?

The set of vectors $V = \{(a, b) \in \mathbb{R}^2\}$ with scalar multiplication defined by $k(a, b) = (ka, b)$

Vector Subspace:

Suppose that V and W are two vector spaces that have identical definitions of vector addition and scalar multiplication, and that W is a subset of V i.e. $W \subseteq V$.

Then W is a **subspace** of V . A vector subspace should also satisfy all the properties of a vector space.

Testing Subsets for Subspaces

Suppose that V is a vector space and W is a subset of V , $W \subseteq V$. Endow W with the same operations as V . Then W is a subspace if and only if three conditions are met

1. **W is non-empty i.e. $W \neq \emptyset$.**
2. **If $\vec{x} \in W$ and $\vec{y} \in W$, then $\vec{x} + \vec{y} \in W$.**
3. **If $\alpha \in \mathbb{R}$ and $\vec{x} \in W$, then $\alpha\vec{x} \in W$.**

This can be paraphrased by saying that a subspace is “a nonempty subset (of a vector space) that is closed under vector addition and scalar multiplication.”

Let's look at an example:

$V = \{(a, b) \in \mathbb{R}^2\}$ with scalar multiplication
defined by $k(a, b) = (ka, b)$

we need to check if our new scalar multiplication
is closed and satisfies all 4 axioms of
validity for scalar multiplication for a vector
space:

→ Checking for closedness
If $\begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}^2$ then $k \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} ka \\ b \end{bmatrix} \in \mathbb{R}^2$

∴ closed

→ $a(\vec{x} + \vec{y}) = a\vec{x} + a\vec{y}$

Let $\vec{x} = \begin{bmatrix} a_1 \\ b_1 \end{bmatrix}$, $\vec{y} = \begin{bmatrix} a_2 \\ b_2 \end{bmatrix}$

$$a(\vec{x} + \vec{y}) = a \begin{bmatrix} a_1 + a_2 \\ b_1 + b_2 \end{bmatrix} = \begin{bmatrix} a(a_1 + a_2) \\ b_1 + b_2 \end{bmatrix}$$
$$a\vec{x} + a\vec{y} = a \begin{bmatrix} a_1 \\ b_1 \end{bmatrix} + a \begin{bmatrix} a_2 \\ b_2 \end{bmatrix}$$
$$= \begin{bmatrix} aa_1 \\ b_1 \end{bmatrix} + \begin{bmatrix} aa_2 \\ b_2 \end{bmatrix} = \begin{bmatrix} a(a_1 + a_2) \\ b_1 + b_2 \end{bmatrix}$$

∴ LHS = RHS

$$(a+b)\vec{x} = a\vec{x} + b\vec{x}$$

$$\text{Let } \vec{x} = \begin{bmatrix} a_1 \\ b_1 \end{bmatrix}$$

$$(a+b)\vec{x} = (a+b) \begin{bmatrix} a_1 \\ b_1 \end{bmatrix} = \begin{bmatrix} (a+b)a_1 \\ b_1 \end{bmatrix}$$

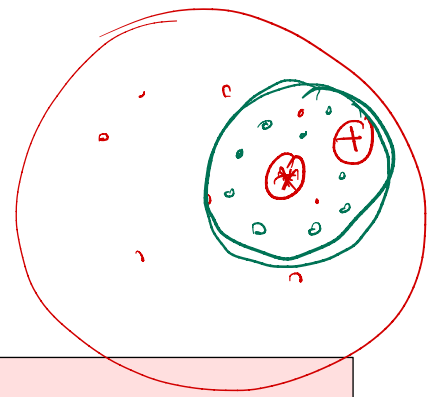
$$\begin{aligned} a\vec{x} + b\vec{x} &= a \begin{bmatrix} a_1 \\ b_1 \end{bmatrix} + b \begin{bmatrix} a_1 \\ b_1 \end{bmatrix} = \begin{bmatrix} aa_1 \\ b_1 \end{bmatrix} + \begin{bmatrix} ba_1 \\ b_1 \end{bmatrix} \\ &= \begin{bmatrix} (a+b)a_1 \\ 2b_1 \end{bmatrix} \end{aligned}$$

$$\therefore \text{LHS} \neq \text{RHS}$$

$\therefore (a+b)\vec{x} \neq a\vec{x} + b\vec{x}$
 $\therefore$ NOT a valid vector space



Hence, this is not a valid subspace!



Let's Practice

Is V a valid Vector Space?

The set of vectors $V = \{(a, b) \in \mathbb{R}^2\}$ with scalar multiplication defined by $k(a, b) = (ka, b)$

Vector Subspace:

Suppose that V and W are two vector spaces that have identical definitions of vector addition and scalar multiplication, and that W is a subset of V i.e. $W \subseteq V$.

Then W is a **subspace** of V . A vector subspace should also satisfy all the properties of a vector space.

Testing Subsets for Subspaces

Suppose that V is a vector space and W is a subset of $V, W \subseteq V$. Endow W with the same operations as V . Then W is a subspace if and only if three conditions are met

1. W is non-empty i.e. $W \neq \emptyset$.
2. If $\vec{x} \in W$ and $\vec{y} \in W$, then $\vec{x} + \vec{y} \in W$.
3. If $\alpha \in \mathbb{R}$ and $\vec{x} \in W$, then $\alpha\vec{x} \in W$.

presence of 0 vector

This can be paraphrased by saying that a subspace is "a nonempty subset (of a vector space) that is closed under vector addition and scalar multiplication."

Let's look at an example:

V is a vector space and S is a subset of V . Determine whether S is a valid subspace of V :

$$V = \mathbb{R}^3$$

$$S = \left\{ \begin{bmatrix} x \\ 12 \\ 3x \end{bmatrix} : x \in \mathbb{R} \right\}$$

Handwritten notes:

$$\vec{a} = \begin{bmatrix} x_1 \\ 12 \\ 3x_1 \end{bmatrix}$$

$$\vec{b} = \begin{bmatrix} x_2 \\ 12 \\ 3x_2 \end{bmatrix}$$

$$\vec{a} + \vec{b} = \begin{bmatrix} x_1 + x_2 \\ 12 + 12 \\ 3x_1 + 3x_2 \end{bmatrix} = \begin{bmatrix} (x_1 + x_2) \\ 24 \\ 3(x_1 + x_2) \end{bmatrix}$$

Let us see what happens when we scale a vector $\vec{x} \in S$ by some $a \in \mathbb{R}$.

$$\text{Let } a = 2$$

$$\text{Let } \vec{x} = \begin{bmatrix} 1 \\ 12 \\ 3 \end{bmatrix} \in S$$

$$\text{Then } a\vec{x} = 2 \begin{bmatrix} 1 \\ 12 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 24 \\ 6 \end{bmatrix} \notin S$$

Hence, S is not closed under scalar multiplication, (the third condition is not satisfied), and thus, S is not a valid subspace of V .

Let's practice:

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Handwritten notes:

$$x=0, y=0 \implies 0-0=12 \times$$

$$3x-5y=12$$

V is a vector space and S is a subset of V . Determine whether S is a valid subspace of V :

$$V = \mathbb{R}^2$$

$$S = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : 3x - 5y = 12 \right\}$$

$$3(x_1 + x_2) - 5(y_1 + y_2) = 12 + 12 = 24$$

Handwritten notes:

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \quad \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

$$V = \mathbb{R}^2$$

$$S = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : 3x - 5y = 12 \right\}$$

Checking for closure under vector addition

$$\vec{a}_1 = \begin{bmatrix} x_1 \\ \frac{3x_1 - 12}{5} \end{bmatrix}$$

$$\vec{a}_2 = \begin{bmatrix} x_2 \\ \frac{3x_2 - 12}{5} \end{bmatrix}$$

$$\vec{a}_1 + \vec{a}_2 = \begin{bmatrix} x_1 + x_2 \\ \frac{3x_1 - 12}{5} + \frac{3x_2 - 12}{5} \end{bmatrix}$$

$$= \begin{bmatrix} (x_1 + x_2) \\ \frac{3(x_1 + x_2) - 24}{5} \end{bmatrix} \notin S$$

$\frac{3x - 24}{5}$

∴ NOT
∴ NOT

closed under vector addition
a valid subspace

Alternatively,

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} \notin S$$

since $3(0) - 5(0) \neq 12$

$$\vec{y} =$$

$$a_1 \vec{x}_1 + a_2 \vec{x}_2 + a_3 \vec{x}_3 \dots a_n \vec{x}_n$$

Linear Combination

A vector $\vec{y}$ in vector space V (over $\mathbb{R}$) is a linear combination of the vectors $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n$ if there exist scalars $a_1, a_2, \dots, a_n$ in $\mathbb{R}$ such that $\vec{y} = a_1 \vec{x}_1 + a_2 \vec{x}_2 + \dots + a_n \vec{x}_n$

Example: The vector $(2,3)$ in $\mathbb{R}^2$ is a linear combination of the vectors $(1,0)$ and $(0,1)$ since $(2,3) = 2(1,0) + 3(0,1)$.

Let's practice:

Prove that the vector $(4,5)$ is a linear combination of the vectors $(3, 1)$ and $(1, 4)$

Will the vectors $(3,1)$ and $(1,4)$ generate any vector we choose in $\mathbb{R}^2$?

To check this, we need to see if, for any vector, say, (b_1, b_2) in $\mathbb{R}^2$, we can find scalars a_1 and a_2 such that $a_1(3,1) + a_2(1,4) = (b_1, b_2)$. This is the same as solving the following system of equations:

$$3a_1 + a_2 = b_1$$

$$a_1 + 4a_2 = b_2$$

and this should always have a solution for a_1 and a_2 . How do we do that?

$$a_1 \begin{bmatrix} 3 \\ 1 \end{bmatrix} + a_2 \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

→ solving for a_1 and a_2 :

$$\begin{aligned} & \left[\begin{array}{cc|c} 3 & 1 & 4 \\ 1 & 4 & 5 \end{array} \right] \xrightarrow[R_1 = R_1 - 3R_2]{R_2 = R_2 - 4R_1} \left[\begin{array}{cc|c} 0 & 1 & 1 \\ 1 & 4 & 5 \end{array} \right] \\ & \xrightarrow{R_1 \rightleftharpoons R_2} \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \end{array} \right] \end{aligned}$$

$$\therefore a_1 = 1$$

$$a_2 = 1$$

$\Rightarrow \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ is a linear combination of $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$

Hint: Think about Non-Singular Matrices!

We can now formalize this situation into our next concept.

Vector Space Generating Set

In linear algebra, the concept of "*span*" is fundamental and helps us understand how sets of vectors can generate entire spaces. *The **span** of a set of vectors is defined as the collection of all possible linear combinations of those vectors.*

Formally speaking,

*If you have a set of vectors, $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n$, their **span** is the collection of vectors that can be expressed in the form:*

$$a_1\vec{x}_1 + a_2\vec{x}_2 + \dots + a_n\vec{x}_n$$

Where $a_1, a_2, \dots, a_n$ are scalars (real numbers). Essentially, the span of these vectors is the set of all vectors that can be formed by scaling and adding the original vectors.

When a set of vectors spans an entire Vector Space, then it is called a Vector Space Generating Set.

Let's take an example:

Consider the vectors $\vec{x}_1 = (3,1)$ and $\vec{x}_2 = (1,4)$. Then $a_1\vec{x}_1 + a_2\vec{x}_2$ can be written as:

$$a_1(3,1) + a_2(1,4)$$

Then the span of $\vec{x}_1$ and $\vec{x}_2$ can be written as :

$$\text{Span} \{ \vec{x}_1, \vec{x}_2 \} = \left\{ \begin{pmatrix} 3a_1 + a_2 \\ a_1 + 4a_2 \end{pmatrix} \mid a_1, a_2 \in \mathbb{R} \right\}$$

Since

$$3a_1 + a_2 = b_1$$

$$a_1 + 4a_2 = b_2$$

will always have a solution, $\vec{x}_1$ and $\vec{x}_2$ generate all possible vectors in $\mathbb{R}^2$ and thus they span the entire $\mathbb{R}^2$ vector space!

Linear Independence

If a vector can be constructed from a linear combination of the others, the set is

“linearly dependent.” If a set of vectors is linearly independent, removing a vector should shrink the span, since that vector itself can't be reconstructed by scaling and

$$\left\{ \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \end{bmatrix} \right\}$$
$$\left\{ \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \end{bmatrix} \right\}$$

adding the other vectors. (If it could, then it wouldn't be linearly independent.)

Similarly, if a vector not in the span is added to the set, then the span should grow.

Formally,

A set of vectors $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n\}$ from a real vector space V is linearly independent

if the only solution to the equation $a_1\vec{x}_1 + a_2\vec{x}_2 + \dots + a_n\vec{x}_n = \vec{0}$ is

$a_1 = a_2 = a_3 = \dots = a_n = 0$. Otherwise, the set is called a linearly

dependent set.

$$\begin{bmatrix} 5 & 2 & 4 \\ -2 & -3 & 5 \\ 4 & 5 & -7 \end{bmatrix}$$

Let's take an example:

We want to check whether the vectors $(5, -2, 4)$, $(2, -3, 5)$, and $(4, 5, -7)$ are

linearly independent or dependent.

$$a_1 \begin{bmatrix} 5 \\ -2 \\ 4 \end{bmatrix} + a_2 \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix} + a_3 \begin{bmatrix} 4 \\ 5 \\ -7 \end{bmatrix}$$

Let's denote the vectors by $\vec{u} = (5, -2, 4)$, $\vec{v} = (2, -3, 5)$, and $\vec{w} = (4, 5, -7)$.

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Let $a_1\vec{u} + a_2\vec{v} + a_3\vec{w} = \vec{0}$

$$\left[\begin{array}{ccc|c} 5 & 2 & 4 & 0 \\ -2 & -3 & 5 & 0 \\ 4 & 5 & -7 & 0 \end{array} \right]$$

For the vectors to be linearly independent, the only solution to the above equation

is: $a_1 = a_2 = a_3 = 0$. In matrix form, the equation would look like:

$$a_1 \begin{bmatrix} 5 \\ -2 \\ 4 \end{bmatrix} + a_2 \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix} + a_3 \begin{bmatrix} 4 \\ 5 \\ -7 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Can we solve this system of equations to check if we can get unique solutions or not? If it does have a unique solution, what will it be? If **not**, are they linearly dependent?

$$\left\{ \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \end{bmatrix} \right\}$$

Let's Practice:

- Is the following set of vectors linearly independent?

$$1. \quad \vec{u} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, \vec{v} = \begin{bmatrix} -2 \\ -8 \end{bmatrix}$$

$$2. \quad \vec{u} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}, \vec{v} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}, \vec{w} = \begin{bmatrix} -4 \\ 7 \end{bmatrix}$$

- Is the vector $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ in the span of set $T = \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 4 \end{bmatrix}, \begin{bmatrix} -2 \\ 2 \\ 3 \end{bmatrix} \right\}$?

$$\left\{ \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \end{bmatrix} \right\}$$

↓

Basis

A basis for a vector space is a set of vectors that is linearly independent and generated the vector space.

Let V be a subspace of $\mathbb{R}^n$. A **basis** of V is a set of vectors $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m\}$ in V such that:

- $V = \text{Span}\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m\}$, and
- The set $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m\}$ is linearly independent.

$$1. a) \begin{bmatrix} -1 \\ 4 \end{bmatrix} \quad \begin{bmatrix} -2 \\ -8 \end{bmatrix}$$

→ Let a_1 and a_2 be 2 scalars $\in \mathbb{R}$

$$a_1 \begin{bmatrix} -1 \\ 4 \end{bmatrix} + a_2 \begin{bmatrix} -2 \\ -8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Solving for a_1 and a_2

$$\begin{bmatrix} -1 & -2 & | & 0 \\ 4 & -8 & | & 0 \end{bmatrix} \xrightarrow{R_1 = -R_1} \begin{bmatrix} 1 & 2 & | & 0 \\ 4 & -8 & | & 0 \end{bmatrix}$$

$$\xrightarrow{R_2 = R_2 + 4R_1} \begin{bmatrix} 1 & 2 & | & 0 \\ 8 & 0 & | & 0 \end{bmatrix} \xrightarrow{R_2 = R_2/8} \begin{bmatrix} 1 & 2 & | & 0 \\ 1 & 0 & | & 0 \end{bmatrix}$$

$$\xrightarrow{R_1 \rightleftharpoons R_2} \begin{bmatrix} 1 & 0 & | & 0 \\ 1 & 2 & | & 0 \end{bmatrix} \xrightarrow{R_2 = R_2 - R_1} \begin{bmatrix} 1 & 0 & | & 0 \\ 0 & 2 & | & 0 \end{bmatrix}$$

$$\xrightarrow{R_2 = R_2/2} \begin{bmatrix} 1 & 0 & | & 0 \\ 0 & 1 & | & 0 \end{bmatrix} \quad \therefore a_1 = 0$$

$$a_2 = 0$$

$\therefore$ linearly independent

$$b) a_1 \begin{bmatrix} 3 \\ -1 \end{bmatrix} + a_2 \begin{bmatrix} 4 \\ 5 \end{bmatrix} + a_3 \begin{bmatrix} -4 \\ 7 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 3 & 4 & -4 & 0 \\ -1 & 5 & 7 & 0 \end{array} \right]$$

Here we have 3 unknowns and 2 equations

$\therefore$ we will have infinitely many solutions.

$\therefore$ linearly dependent

whenever we have more vectors in a set than the dimensionality of the vectors, it is a linearly dependent set.

2.

$$a_1 \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} + a_2 \begin{bmatrix} 0 \\ 1 \\ 4 \end{bmatrix} + a_3 \begin{bmatrix} -2 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 2 & 0 & -2 & 1 \\ -1 & 1 & 2 & 1 \\ 1 & 4 & 3 & 1 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} -1 & 1 & 2 & 1 \\ 2 & 0 & -2 & 1 \\ 1 & 4 & 3 & 1 \end{array} \right]$$

$$\begin{array}{l} R_2 = R_2 + 2R_1 \\ R_3 = R_3 + R_1 \end{array} \rightarrow \left[\begin{array}{ccc|c} -1 & 1 & 2 & 1 \\ 0 & 2 & 2 & 3 \\ 0 & 5 & 5 & 2 \end{array} \right] \begin{array}{l} R_2 = R_2 / 2 \\ R_3 = R_3 / 5 \end{array}$$

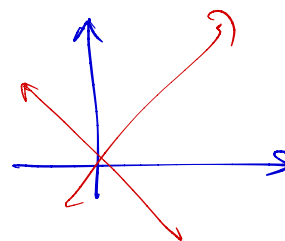
$$\left[\begin{array}{ccc|c} -1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 3/2 \\ 0 & 1 & 1 & 2/5 \end{array} \right] \xrightarrow{R_3 = R_3 - R_2} \left[\begin{array}{ccc|c} -1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 3/2 \\ 0 & 0 & 0 & -11/10 \end{array} \right]$$

$\therefore$ No solution
 $\therefore$ Not in span

Recall that a set of vectors is *linearly independent* if and only if, when you remove any vector from the set, the span shrinks. In other words, if $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m\}$ is a basis of a subspace V , then no proper subset of $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m\}$ will span V : it is a **minimal spanning set**.

The **dimension** of a space is the number of basis vectors required to span it. For a space like $\mathbb{R}^n$, it should be intuitive that this number is n , and we couldn't somehow span an n -dimensional space with fewer vectors than that.

Let's look at an example:



Let's try to look at a basis for $\mathbb{R}^2$:

We need to have 2 linearly independent vectors in $\mathbb{R}^2$ that generate $\mathbb{R}^2$. One such basis can be $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$. Why?

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

1. Any vector $\begin{bmatrix} a \\ b \end{bmatrix}$ can be written as a linear combination of $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$:

$$a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

2. They are linearly independent:

$$x \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ gives } x = 0 \text{ and } y = 0$$

↓
Basis vectors
for $V = \mathbb{R}^2$
 $\left\{ \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \end{bmatrix} \right\} \rightarrow$ Basis vectors
for $V = \mathbb{R}^2$

$\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ is the standard basis for $\mathbb{R}^2$

$$\left\{ \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$$

Generally, the standard co-ordinate vectors:

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, \vec{e}_n = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} \text{ form a basis for } \mathbb{R}^n. \text{ This is known as}$$

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

the *standard basis*.

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Change of Basis

A vector space can have multiple bases and not all will be standard basis, but can still generate the entire vector space. We can think of it as defining a coordinate system, which isn't necessarily aligned with the original xyz axes, and doesn't even guarantee that the axes of the coordinate system are at right angles.

Let's look at an example:

Let's look at the following basis for $\mathbb{R}^2$:

$\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \end{bmatrix} \right\}$ (we can check and see that they span entire $\mathbb{R}^2$ and are linearly independent).

$$\begin{bmatrix} x' \\ y' \end{bmatrix}$$

$$3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 5 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = x' \begin{bmatrix} 1 \\ 1 \end{bmatrix} + y' \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

Now a vector $\begin{bmatrix} 3 \\ 5 \end{bmatrix}$ in standard co-ordinate system will have different co-ordinates in this new basis (new co-ordinate system):

$$3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 5 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

In general, if we are given vectors in one co-ordinate system, it is useful to know how to find the co-ordinates of the vector in another co-ordinate system:

If we have two bases:

1. $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m\}$ and

2. $\{\vec{y}_1, \vec{y}_2, \dots, \vec{y}_m\}$

and we are given the co-ordinates of a vector $\vec{v}$ in basis 1 and we want to find the co-ordinates of $\vec{v}$, say $\vec{v}'$, in basis 2, we use:

$$\underline{v_1 \vec{x}_1} + \underline{v_2 \vec{x}_2} + \dots + v_m \vec{x}_m = v'_1 \vec{y}_1 + \underline{v'_2 \vec{y}_2} + \dots + \underline{v'_m \vec{y}_m}$$

Sometimes you just want to describe some data in a different way. Personality tests, for example, could describe a person's personality in many different ways, and a change of basis could turn test results that use one description (studiousness, extraversion, etc) into another set of traits (perhaps leadership involves both studiousness and extraversion).

Another application of a change of basis is color. Many programs use values of Red, Green, and Blue (RGB) as a way of specifying a color, but there's nothing inherently special about that basis. The color schemes YUV and YCbCr both represent color as a luminance component, Y, that describes the overall brightness, followed by two color components. When streaming a movie, it can be helpful for compression to represent the color in YUV or YCbCr because it can downsample the color components while keeping the resolution of the luminance channel. (The human eye perceives color at less resolution than brightness.) Transforming RGB into YUV or YCbCr takes just a matrix multiplication, as does changing back from those bases.

Let's Practice

- Find the co-ordinates for vector $\vec{w}$ (in standard basis) relative to the basis $S = \{\vec{u}_1, \vec{u}_2\}$ for $\mathbb{R}^2$:

$$\vec{w} = (3, -7)$$

$$\vec{u}_1 = (2, -4)$$

$$\vec{u}_2 = (3, 8)$$

- Does this set of vectors form the basis for $\mathbb{R}^3$?

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix} \right\}$$

1. Let the co-ordinates of $\vec{w}$ in new basis be $\begin{bmatrix} x \\ y \end{bmatrix}$

$$3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + (-7) \begin{bmatrix} 0 \\ 1 \end{bmatrix} = x \begin{bmatrix} 2 \\ -4 \end{bmatrix} + y \begin{bmatrix} 3 \\ 8 \end{bmatrix}$$

solving for x and y

$$\left[\begin{array}{cc|c} 2 & 3 & 3 \\ -4 & 8 & -7 \end{array} \right] \xrightarrow{R_2 = R_2/2} \left[\begin{array}{cc|c} 2 & 3 & 3 \\ -2 & 4 & -7/2 \end{array} \right]$$

$$\begin{array}{l} R_1 = R_1 + R_2 \\ R_2 = R_2/2 \end{array} \rightarrow \left[\begin{array}{cc|c} 0 & 7 & 7/2 \\ -1 & 2 & -7/4 \end{array} \right] \xrightarrow{R_1 = R_1/7} \left[\begin{array}{cc|c} 0 & 1 & -1/4 \\ -1 & 2 & -7/4 \end{array} \right]$$

$$\begin{array}{l} R_1 \leftrightarrow R_2 \\ R_1 = -R_1 \end{array} \rightarrow \left[\begin{array}{cc|c} 1 & -2 & 7/4 \\ 0 & 1 & -1/4 \end{array} \right] \xrightarrow{R_1 = R_1 + 2R_2} \left[\begin{array}{cc|c} 1 & 0 & 45/28 \\ 0 & 1 & -1/4 \end{array} \right]$$

$$\left[\begin{array}{cc|c} 1 & 0 & 45/28 \\ 0 & 1 & -1/4 \end{array} \right]$$

$\therefore$ The new co-ordinates

are $\begin{bmatrix} 45/28 \\ -1/4 \end{bmatrix}$

2.

$$\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix}$$

In order for $\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix} \right\}$ to span $\mathbb{R}^3$,

they should be linearly independent and

generate all 3D real vectors.

For that
$$a_1 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + a_2 \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

should always lead to unique solutions

i.e.
$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \\ 0 & 1 \end{bmatrix}$$
 should reduce to an

identity matrix which is NOT possible

$\therefore$ NOT a Basis

If the number of vectors is less than the dimension of the vector space, they cannot generate the vector space.